

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA0301

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO4:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Mock Interview

Duration: 1hr

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment.

Answer:

Reinforcement Learning (RL) is a learning method in which an **agent learns by interacting with an environment**. At each step, the agent observes the current **state**, selects an **action**, and receives a **reward** from the environment. The agent then moves to a new state.

The learning process is:

State → Action → Reward → New State → Policy Update

The agent tries different actions through **exploration** and learns to select the best actions through **exploitation**. Over repeated interactions, it updates its policy or value function to maximize the **cumulative future reward**.

For example, a robot receives positive rewards for reaching a destination and negative rewards for collisions. After many trials, it learns an optimal path.

2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process.

Answer:

I would formulate the ride-sharing or delivery problem as a **Markov Decision Process (MDP)**. An MDP represents decision-making using **states, actions, rewards, transitions, and policy**.

- **State:** Driver location, traffic, available vehicles, customer requests, delivery status, and time.
- **Action:** Assign a driver, select a route, accept a request, or dispatch a vehicle.

- **Reward:** Positive reward for successful and fast delivery; penalties for delays, fuel consumption, and cancellations.
- **Transition:** The system changes to a new state after an action.
- **Policy:** Determines the best action for each state.

The objective is to maximize long-term reward while reducing **delivery time, operating cost, and customer waiting time.**

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty.

Answer:

Model-Free RL learns directly from interaction with the environment without explicitly learning how the environment works. Algorithms such as **DQN and PPO** learn a policy or value function from collected experiences.

Model-Based RL learns or uses a model of the environment. The model predicts the next state and reward for a particular action. The agent can then use this model to plan future actions.

Feature	Model-Free RL	Model-Based RL
Environment model	Not required	Required/learned
Planning	Limited	Strong
Sample efficiency	Usually lower	Usually higher
Examples	DQN, PPO	MPC, World Models
Uncertainty	Learned from experience	Can model/predict uncertainty

Model-Based RL can handle uncertainty by predicting possible future states, while Model-Free RL learns directly from repeated environmental experiences.

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved.

Answer:

Reward sparsity occurs when the agent receives rewards only occasionally. For example, a recommendation system may receive a reward only when a user purchases an item. This makes it difficult for the agent to understand which recommendation was successful.

To improve reward sparsity, I would:

- Introduce meaningful intermediate rewards such as clicks, likes, or watch time.
- Improve exploration.
- Use suitable reward shaping.
- Analyze user interaction data.

For **hyperparameter tuning**, I would test parameters such as:

- Learning rate
- Discount factor
- Exploration rate
- Batch size
- Network size

I would compare different settings using **average reward, convergence speed, and recommendation performance** to select the best configuration.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability.

Answer:

I would design the RL game-playing system using the following pipeline:

Game State → Agent → Action → Game Environment → Reward → Experience → Training → Updated Agent

The agent observes the game state and selects an action. The game provides a reward based on the result. The experience is stored and used to update the model.

To improve training stability, I would use:

- **Experience replay** to reuse previous experiences.
- **Target networks** for stable value estimation.
- Appropriate learning rates.
- Reward normalization or clipping.
- Controlled exploration such as ϵ -greedy.
- Regular evaluation and model checkpoints.

I would monitor **reward, loss, and win rate** to determine whether the AI is learning correctly.

6. A company wants to use Reinforcement Learning for dynamic pricing. Explain how the agent learns from interaction and improves pricing decisions over time.

Answer:

In dynamic pricing, the RL agent learns the best price by observing market conditions and customer responses.

The **state** may include demand, inventory, competitor prices, customer behavior, and time. The **action** is the price selected by the agent. The **reward** can represent profit, sales, and customer satisfaction.

The process is:

Observe Market → Select Price → Observe Customer Response → Receive Reward → Update Policy

For example, if the agent sets a price that increases profit without significantly reducing demand, it receives a positive reward. Over repeated interactions, it learns which pricing strategies provide higher long-term returns.

7. If an RL model is not converging properly, explain how you would identify and fix issues related to learning rate and exploration strategy.

Answer:

First, I would analyze the **reward curve, loss curve, and policy performance** to identify whether the model is unstable or learning too slowly.

Learning Rate:

- If the learning rate is **too high**, the model may make very large updates and fail to converge.
- If it is **too low**, learning may be extremely slow.

I would test different learning rates and use learning-rate scheduling if necessary.

Exploration:

If exploration is too high, the agent keeps taking random actions. If exploration is too low, the agent may stop exploring and get stuck in a poor policy.

A common solution is to gradually reduce exploration:

High Exploration → Gradual Reduction → Increased Exploitation

This allows the agent to discover good actions initially and use them more frequently later.

8. A robotics system uses Reinforcement Learning for navigation. Explain how the environment and agent interaction leads to optimal path learning.

Answer:

In robotic navigation, the **robot is the agent** and the physical or simulated surroundings represent the environment.

The state may contain:

- Robot position
- Distance to destination
- Obstacle locations
- Sensor readings
- Current direction

The actions may include **forward, backward, left, right, or stop**.

The robot receives:

- **Positive reward** for moving toward the destination or reaching it.
- **Negative reward** for collisions, unnecessary movements, or taking long paths.

The learning cycle is:

Observe → Select Action → Move → Receive Reward → Update Policy

After many trials, the robot learns actions that maximize reward and eventually discovers a safe and efficient path.

9. If you are asked to improve an RL model with delayed rewards, explain techniques to handle credit assignment problems.

Answer:

A **delayed reward** occurs when the agent receives a reward much later than the action that caused it. This creates a **credit assignment problem**, because the agent may not know which previous action contributed to the final result.

To solve this problem, I would use:

- **Discount factor:** Helps connect current actions with future rewards.
- **n-step returns:** Uses rewards from multiple future steps.
- **Eligibility traces:** Keeps information about recently visited states and actions.
- **Advantage estimation:** Identifies actions that performed better or worse than expected.
- **Reward shaping:** Provides intermediate rewards when appropriate.

These methods help the agent understand the contribution of earlier actions and improve long-term decision-making.

10. A financial firm uses Reinforcement Learning for trading decisions. Explain how uncertainty and risk can be handled in the learning process.

Answer:

Financial markets are uncertain because prices can change due to economic conditions, news, market trends, and unexpected events. Therefore, an RL trading agent should consider both **profit and risk**.

The state can include:

- Historical prices
- Trading volume
- Market indicators
- Portfolio balance
- Market volatility

The actions can be **buy, sell, or hold**.

The reward should consider risk as well as return:

Reward = Investment Return – Risk Penalty – Transaction Cost

Risk can be controlled using **portfolio limits, conservative exploration, risk-sensitive rewards, and maximum drawdown constraints**. The model should also be tested using historical data and simulations before real-world deployment.

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately